

What is the biochemical and physiological rationale for using cold-water immersion in sports recovery? A systematic review

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ABSTRACT

Cold-water immersion (CWI) is a popular recovery intervention after exercise. The scientific rationale is not clear, and there are no clear guidelines for its use. The aim of this review was to study the physiological and biochemical effect of short periods of CWI. A computer-based literature search, citation tracking and related articles searches were undertaken. Primary research studies using healthy human participants, immersed in cold water (<15°C), for 5 min durations or less were included. Data were extracted on body temperature, cardiovascular, respiratory and biochemical response. 16 studies were included. Sample size was restricted, and there was a large degree of study heterogeneity. CWI was associated with an increase in heart rate, blood pressure, respiratory minute volume and metabolism. Decreases in end tidal carbon dioxide partial pressure and a decrease in cerebral blood flow were also reported. There was evidence of increases in peripheral catecholamine concentration, oxidative stress and a possible increase in free-radical-species formation. The magnitude of these responses may be attenuated with acclimatisation. CWI induces significant physiological and biochemical changes to the body. Much of this evidence is derived from full body immersions using resting healthy participants. The physiological and biochemical rationale for using short periods of CWI in sports recovery still remains unclear.

INTRODUCTION

It is commonplace for athletes to employ postexercise strategies or practices in a bid to speed up recovery. Recently, cold-water immersion (CWI) has emerged as one of the most popular interventions. Despite its popularity, the evidence from clinical trials remains equivocal,^{1–7} and there is little evidence to suggest an optimal treatment protocol.⁸

Currently, most athletes and sports practitioners follow anecdotal guidelines. High-level athletes in Australia may favour three intermittent cycles of 1 min immersions,⁵ whereas others speculate that clinical durations are as short as 30 s.⁹ Furthermore, there is little scientific rationale on what the lower limit for water temperature should be, and athlete tolerance or preference is often the basis. CWI has been defined as water temperatures of less than 15°C,⁹ based on the temperature at which cold pain begins;¹⁰ however, primary research studies have employed lower temperatures between 8°C and 10°C,^{2,6} with others lowering water to 5°C.⁵

Few have considered the basic scientific rationale underpinning the use of CWI in sports recovery. It is clear that CWI induces a degree of shock on the body, causing a number of physiological and biochemical responses, but the subsequent effect on recovery is not clear. Perhaps the most common concept is that CWI induces vasoconstriction, stimulating venous return and aiding metabolite removal after exercise; reducing swelling and muscle soreness.⁸ In contrast, other systemic effects associated with CWI are often overlooked in the sporting context. Other fields of research^{11–14} have examined the body's response to extreme cold temperatures in terms of adaptation, biochemical status, immunology, respiration and cardiovascular response, but this has not yet been extrapolated to a sports recovery model.

Continued disparity in the rationale for CWI and vague guidelines for its use after sport mean that athletes could risk employing more extreme temperatures or longer immersion times, before determining actual benefit or risk. The aim of this review is to determine the biochemical and physiological effect of CWI. The objectives are:

1. To study the effects of short periods of CWI (<15°C) on body temperature, cardiovascular system, respiratory system and biochemical responses in healthy human participants.
2. To relate these effects to athletic recovery by determining any potential beneficial or harmful effects.

METHODS

Search strategy

Two independent researchers undertook a computerised literature search in February 2007 (<http://www.pubmed.gov>) using a combination of (Boolean logic [AND]) eight key words or phrases (Ice, 'ice bath', 'cold water immersion', cryotherapy, sports medicine, recovery, temperature, blood flow). This was followed up with a more sensitive search strategy on five databases (British Nursing index (BNI), Cumulative Index to Nursing and Allied Health (CINAHL), EMBASE, MEDLINE and Sports Discus on OVID) up to June 2008 (table 1). This was complemented with citation tracking of key primary and review articles (n=109). All searches were carried out independently, using predetermined inclusion criteria and extraction forms. Details on the titles read, abstracts read, full text articles retrieved, and the excluded and included studies were compared for each author, with any disparities resolved by consensus discussion.

Table 1 Details of search strategyFebruary 2008 (<http://www.pubmed.gov>)

1. Cold water immersion (1145)
2. Ice bath (299)
3. Ice AND sports medicine (186)
4. Cryotherapy AND sports medicine (48)
5. Ice immersion AND recovery (39)
6. Cryotherapy AND sports recovery (19)
7. Ice immersion AND temperature (224)
8. Ice immersion AND blood flow (50)

Titles screened: 2010

Abstracts read: 274

June 2008 (British Nursing index (BNI), Cumulative Index to Nursing and Allied Health (CINAHL), EMBASE, MEDLINE, and Sports Discus on OVID)

1. Ice immersion.mp (41)
2. Water immersion.mp. (5085)
3. cryotherapy.mp. (14376)
4. Oxidative stress.mp. (122761)
5. antioxidants.mp. (152513)
6. biochemistry.mp (87561)
7. ventilation.mp (176336)
8. respiratory.mp. (709488)
9. cardiovascular.mp (608105)
10. Heart rate.mp. (311469)
11. Blood flow.mp. (334204)
12. Skin temperature.mp. (17476)
13. Core temperature.mp (9314)
14. 1–3 (OR) (19391)
15. 4–13 (OR) (2161219)
16. 14 and 15 (2661)
17. Remove duplicates from 16 (1961)

Titles screened: 1961

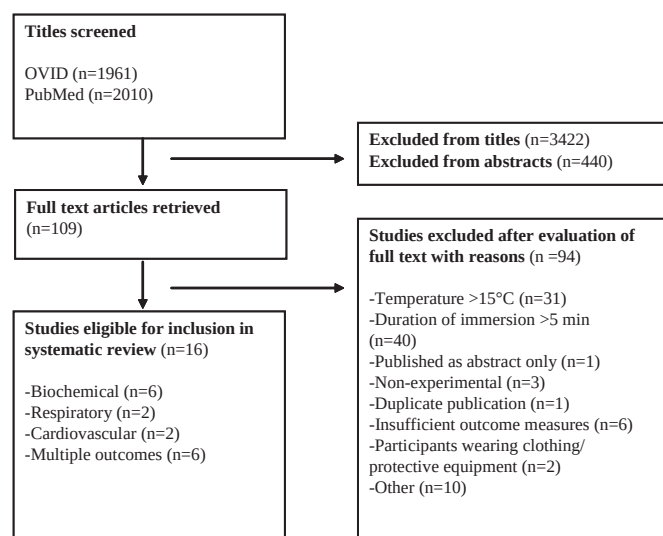
Abstracts read=166

Inclusion criteria

To be included in the review, studies had to fulfil the following criteria: participants should be healthy human participants, with no restrictions placed on their experience of CWI. Intervention should be immersion in water at <15°C for 5 min or less (including single 'plunge' immersions). Immersion must have been to at least waist level, and participants must have been wearing swimming shorts/trunks or no clothing. Comparisons could have been made to any other intervention (including other forms of cryotherapy or no intervention). Outcomes were body temperature (including core temperature, deep tissue temperature and skin temperature), cardiovascular response (including heart rate, blood pressure, blood flow), respiratory response (including minute volume) and biochemical markers. No limits were placed on the timing of the outcomes. Study design could have been any primary research study, including: randomised controlled trials (RCTs), randomised crossover studies, controlled trials (cohort, before/after), observational or cross-sectional. Note that studies using head submersion or isolated immersion of a body part (eg, foot only, face only, arm only), were excluded.

Data extraction

Two researchers used a standardised form to extract the data and other key methodological details. We focused primarily on data showing the immediate response to CWI. Where available, additional data on the effects of long-term exposure

**Figure 1** QUORUM flow diagram summarising study selection for inclusion.

to CWI, and the status of cold acclimatised participants, were extracted.

RESULTS

Included studies

The QUORUM flow diagram summarises the selection process (figure 1). N=16 studies^{15–30} met the inclusion criteria, with the majority employing a small sample size. A summary of each study is detailed in appendix table A1. A range of CWI protocols were employed (table 2). Three studies were RCTs, of which two^{16 30} focused primarily on cold habituation, and one¹⁵ compared the body's response to either CWI or whole body cryotherapy (WBC). The remainder of studies used either cross over, or controlled designs. Three of the controlled studies^{25–27} were limited to comparing outcomes before and after a single CWI, whereas two^{18 19} followed groups of winter swimmers undertaking multiple CWI, over a 3-month period. Two^{26 27} also included a case-control comparison using cold acclimatised participants, and controls with no experience of CWI. There was a large degree of heterogeneity between studies, in terms of study design, intervention and outcomes, and so results were analysed and discussed qualitatively.

CARDIOVASCULAR RESPONSE

Heart rate

Six studies recorded heart rate during CWI; however only three gave details on preimmersion values.^{21 25 28} Both Mantoni *et al*²⁵ and Tipton *et al*²⁸ found that CWI increased heart rate from preimmersion states. Immediate increases from 74 (SD 16) beats per minutes (bpm) to 107 (SD 18) bpm were found after just 30 s of cold immersion.²⁵ Longer immersions at milder temperatures of 10°C resulted in lower mean heart rates of 95, 92, 84 bpm during the first, second and third minutes of immersion, respectively.²⁸ Others^{16 29 30} also reported a trend towards increases in heart rate. In contrast, a small study by Kauppinen²¹ found that heart rates peaked immediately prior to immersion. There was further evidence that small groups of participants undertaking multiple immersions

APPENDIX

Table A1 Summary of included studies

Study (design)	Participants	Intervention(s)	Preimmersion	Response to immersion	Notes
Goode et al ¹⁷ (crossover)	N=8 healthy participants (two were experienced divers); aged 20–34 years	A. 11.1°C B. 28°C All groups Immersion to neck Maximum of 5 min duration Sudden immersion Swimming trunks	Mean Ventilation (l/min) Not provided P.Pco ₂ (mm Hg) A. 36.4; B. Not provided	Mean Ventilation (l/min) 20 s: A. 86.8*** B. 24 P.Pco ₂ (mm Hg) Breath 1: A. 23.9 (1.4); B. 30.7 (2.6) Breath 2: A. 27.5 (1.3); B. 35.8 (1.2) Breath 3: A. 28.4 (1.2); B. 36.2 (1.3) Breath 10: A. 29.4 (1.3); B. 36.7 (1.0)	Small sample size, statistical methods unclear. The sudden nature of the immersions and may have maximised the respiratory responses. Preimmersion data unclear; outcomes limited to the first 10 breaths/20 s of immersion.
Malkinson et al ²⁴ (crossover)	N=116 healthy participants (58 male, 58 female); aged 20–55 years	A. 13.53°C B. 33.18°C All groups Immersion to shoulder level 4 min duration Water turbulence Standard swimming attire	Mean ventilation (l/min) Not provided	Mean ventilation (l/min) 30 s: A. Increase by 520%** (male)/360%** (female) from resting levels B. Increase by 72–76% 4 min: A. Mean increase of 175%*** (male)/75%*** (female) from resting levels Overall: A. 43.68*** (20.6) (males); 28.74*** (10.3) (female)	Preimmersion data unclear. Statistical methods unclear. Small number of participants underwent preheating in a sauna. Significantly** greater increase in ventilation rate in males com- pared with females during CWI.
Kauppinen ^{20–22} (crossover)	N=9 healthy winter swimmers; aged 22–50 years	A. Sauna/CWI B. Sauna/cold shower C. Sauna D. CWI (temperature not specified, immersion to neck level, 30 s × 2 with 2 min rest between)	Heart rate (BPM) D. 93.9 (16.8)* Systolic arterial pressure (mm/Hg) D. 128.7 (15.4) Diastolic blood pressure (mm/Hg) D. 86 (7.1) Haemoglobin concentrations (g/l) D. 144 (5) Plasma volume Not provided Core temperature (°C) D. 37.2 Mean skin temperature (°C) D. 32.9	Heart rate (BPM) CWI 1: 70.3 (18.1) CWI 2: 73.2 (19.2) 30 mins: 54.1 (5.6)* Systolic arterial pressure (mm/Hg) CWI 1: 143.1 (17.2)* CWI 2: 147.3 (14)* 30 min: 124.7 (10.3) Diastolic blood pressure (mm/Hg) CWI 1: 85.3 (6.9) CWI 2: 91.8 (9.7)* 30 min: 85.6 (6.4) Haemoglobin concentrations (g/l) CWI 1: 145 (6) 30 min: 141 (6) Plasma volume (% change) D. Decrease of 1.3 (3)% Core temperature (°C) Overall decrease of 0.1 Mean skin temperature (°C) CWI 1: 30.8	Small sample size. Data extracted for group D only. Note all other groups underwent preheating in sauna. Heart rates were significantly higher pre treatment (baseline) in the CWI group (D) in compari- son with others (A–C). Heart rates significantly lower at 30 min post-treatment in Gp D versus group A–C. Significant* increase in systolic arterial pressure after both immersions versus baseline. Diastolic blood pressure was significantly higher* versus baseline on the second immer- sion only.
Kauppinen et al ²³ (crossover)	N=9 winter swimmers Mean age 34.9 years (SD: 10.7). Winter swimming for 1–18 years	A. Sauna and CWI (head out, 7°C for 10 s) B. Sauna and 15°C shower C. Sauna and room temperature rest D. CWI immersion (head out, 6°C for 20 s) 2 min rest at room temperature and repeat.	Plasma ACTH (ng/l) D. 28.56 (14.97) Serum cortisol (nmol/l) D. 235 (103.1) Serum melatonin (pg/l) D. 7.48 (2.50) Plasma norepinephrine (n/ mol) D. 2.44 (0.57) Plasma epinephrine (n/mol/l) D. 1.0.23 (0.07)	Plasma ACTH (ng/l): Immediately: D. 33.50 (11.78) 30 min: D. 24.17 (10.51) Serum cortisol (nmol/l) Immediately: D. 355.2 (102.5)** 30 min: D. 293.3 (66.4) Serum melatonin (pg/l) Immediately: D. 7.90 (2.45) 30 min: D. 7.83 (1.78) Plasma norepinephrine (n/mol) Immediately: D. 4.10 (0.79)** 30 min: D. 3.62 (0.93) Plasma epinephrine (n/mol/l) Immediately: D. 0.27 (0.06) 30 min: D. 0.28 (0.5)	Small sample size * Depicts significant increases from baseline values. (See key for degree of significance)
Tipton et al ²⁸ (crossover)	N=9 healthy male participants; aged 20–35 years	A. Swimming trunks B. Conventional clothing C. Conventional clothing plus wind proof/shower proof clothing All groups 3 min 10°C stirred water Head out immersion Slowly lowered into water	Minute volume (l/min) A. 16.4 Average oxygen consumption (l/min) A. 0.417 Heart rate (BPM) 5 min pre: A. 75 1 min pre: A. 95 P.Pco ₂ (kPa) Not provided Skin temperature (°C)~ A: 30°C	Minute volume (l/min) 1 min: A. 31 2 min: A. 32 Average oxygen consumption (l/min) During immersion A. 0.676 Heart rate (BPM) 1 min: A. 92 2 min: A. 84 P.Pco ₂ (kPa) Mean reduction of 0.96 kPa Skin temperature (°C)~ 30 s: D. 21°C 1 min: D. 20°C 1.5 min: D. 19°C 2 min: D. 18°C Rate of change in skin temperature (°C/s) 0–30 s 0.301	Small sample size. Cross over design, at least 1 week separat- ing interventions. SD values missing. Participants encouraged to breath hold for as long as possible on initial immersion ~ Approximate data extracted from graph

Continued

Table A1 Continued

Study (design)	Participants	Intervention(s)	Preimmersion	Response to immersion	Notes
Siems et al ²⁶ (repeated-measures/case-control comparison cold acclimatised versus control at baseline)	A. N=10 healthy participants who regularly participate in winter swimming (at least once per week for 5 min in ice cold water, for 1–11 years); mean age: 40.5 years (12.3) B. N=10 healthy participants; no experience of CWI; mean age: 35.6 years (8.8)	A. CWI (n=10) 5 min immersion 2–3°C Air temperature: 1 and 5°C B. No immersion (n=10)	Total glutathione (mM) A. 3.12 mM (0.47) B. 2.42mM (0.52) Oxidised glutathione (GSSG) (mM) Not provided Reduced glutathione (GSH) (mM) A. 2.81 (0.44) B. 2.08 (0.49) Uric acid (mM) A. 310 (126)	Total glutathione (mM) Not provided Oxidised glutathione (GSSG) (mM) A. Increased after immersion in all participants Reduced glutathione (GSH) (mM) 1 h: A. 2.61 mM (SD not provided) Uric Acid (mM) 1 h: A. 141 (69) 24 h: A. Return to baseline values	CWI tests took place outside with external air temperatures of 1–5°C. Cold-water immersion associated with oxidative stress. Author notes that cold immersion could potentially be used as a mechanism for 'body hardening'
Hermanussen et al ¹⁸ (repeated measures)	N=11 winter swimmers (six males; 5 female); aged 24–29 years	CWI: 6.8–2°C, 1.5–10 min depending on mood, head out immersion. CWI Test 1: at start of winter swimming season CWI Test 2: after 2.5 months of winter swimming (end of winter swimming season)	Cortisol (ng/ml) CWI test 1: 99 CWI test 2: 101 Thyroid stimulating hormone (mU/l) CWI test 1: 0.96 CWI test 2: 0.93 Glucose (mg/dl) CWI test 1: 78.56	Cortisol (ng/ml) CWI test 1: 30 min post: 133* 60 min post: 99 CWI test 2: 30 min post: 137* 60 min post: 96 Thyroid stimulating hormone (mU/l) CWI test 1: 30 min post: 1.42** 60 min post: 1.23** CWI test 2: 30 min post: 1.43** 60 min post: 1.17** Glucose (mg/dl) CWI test 1: 30 min post: 90.78** 60 min post: 92.44**	Small sample size. CWI in the sea at temperatures between 6.8 and 2°C CWI did not result in any changes in prolactin; luteinising hormone, follicle-stimulating hormone, growth hormone or glucose at any follow-up, and there were no changes to participants' baseline levels. Resting levels of basal prolactin increased, and serum insulin decreased over study period.
Tipton et al ²⁹ (controlled trial)	N=13 healthy male participants; aged 18–32 years	A. Control (n=5) CWI day 1 and day 5. 10°C, 3 min, stirred water, neck level B. Habituation (n=8) CWI at day 1 and day 5. 10°C, 3 min, stirred water, neck level CWI at day 2, 3 and 4: 15°C, 3 min, stirred water, neck level, twice/day	Mean inspiratory minute volume (l/min) Day 1: A. 15 Day 5: A. 15	Mean inspiratory minute volume (l/min) 0–30 s Day 1: A. 60~; B: 67.6 Day 5: A. 65~; B: 29.5 30–180 s~ Day 1: A. 50; B: 55 Day 5: A. 40; B: 20 Mean heart rate 0–30 s~ Day 1: A. 95; B: 115 Day 5: A. 85; B: 90 30–180 s~ Day 1: A. 110; B: 130 Day 5: A. 110; B: 110	Small sample size. Biphasic response to immersion noted: 0–30 s; 30–180 s. ~Values extracted from graph to nearest 5 l/min Few mean (SD) values reported, preimmersion values unclear, statistical comparisons unclear. Mean rate of change in skin temperature: 0.33°C/s in 15°C; 0.42°C/s in 10°C.
Siems et al ²⁷ (repeated-measures/case-control comparison cold acclimatised vs control at baseline)	A. N=36 winter swimmers (23 male, 13 female); winter swimming for at least 2 years (range 2–11 years); mean age: 36.1 years (SD:11.8) B. N=40 healthy controls (28 men, 12 women); mean age: 36.3 years (SD: 10.4)	A. CWI 5–10 min immersion in ice cold water (1–4°C) B. No immersion	Reduced glutathione (mmol/l) (GSH) A: 2.85 (SE: 0.41) B: 2.32 (SE: 0.41) Oxidised glutathione (GSSG) (μmol/l) A. 59.1 (SE: 13.9) Uric acid (μmol/l) A. 354 (SE: 99) HNE (μmol/l) A. 0.077 (SE: 0.013) Ascorbic acid (μmol/l) A. 49 (SE: 13.2) Catalase (U/mgHb)** A. 950~ B. 700~ Superoxidase dismutase** (U/mgHb) A. 450~ B. 340~	Reduced glutathione (mmol/l) (GSH) 1 h: A. 2.46 (SE: 0.46)* Oxidised glutathione (GSSG) (μmol/l) 1 h: A. 81.9 (SE: 22.1)** Uric acid (μmol/l) 1 h: A. 298 (120) HNE (μmol/l) 1 h: A. 0.097 (0.121)** Ascorbic acid (μmol/l) 1 h: A. 30.9 (10.7)**	CWI undertaken outside in air temperatures between –1 and 5°C ~ Approximate data extracted from graph No significant difference in: ASAT, AP, GT, CK, plasma protein, potassium ions, and glucose between baseline and postimmersion in Group A. No significant difference in Glutathione peroxidase at baseline between group A and B.
Tipton et al ³⁰ (randomised controlled trial)	N=12 healthy participants; aged 18–32 years; no prior experience of cold immersion	A. Control (n=4) CWI at day 1 and day 5. 10°C, 3 min, stirred water, neck level (two immersions per day) B. Habituation (n=8) CWI at day 1 and day 5=10°C, 3 min, stirred water, neck level (two immersions per day) CWI at day 2, 3 and 4=10°C, 3 min, stirred water, neck level (two sets) All groups: Follow-up CWI at 2, 4, 7 and 14 months.	No preimmersion values provided	Inspiratory minute volume (l/min) (30–180 s) Day 1: A. 58.8 (17–93); B: 56.8 (25–78) Day 5: A. 47.3 (15–76); B: 17.5 (8–22) 2 month: A. 65.2 (39–95); B: 29.4 (14–57) 14 month: A. 29.6 (14–52); B: 25.4 (12–55) Heart rate (BPM) (30–180 s) Day 1: A. 104 (87–115); B: 116 (91–138) Day 5: A. 89 (64–100); B: 90 (61–110)	Small sample size. No preimmersion values. Similar physical characteristics between groups at baseline. Biphasic response to immersion noted: 0–30 s; 30–180 s. In group A, minute volume during the later stages of CWI were significantly lower* at month 4, 7 and 14 compared with day 1, 5 and month 2. Heart rate was significantly reduced* from day 5 onwards compared with day 1.

Table A1 Continued

Study (design)	Participants	Intervention(s)	Preimmersion	Response to immersion	Notes
				2 month: A. 96 (78–106); B. 90 (64–103) 14 month: A. 80 (64–100); B. 84 (63–115) Skin temperature (overall) Mean rate of fall of skin temperature overall was 0.42°C/s	In group B, minute volume and heart rate during the later stages of CWI, remained significantly lower at all follow-ups* in comparison with day 1.
Huttunen et al ¹⁹ (repeated measures)	N=39 winter swimmers; 7 male; mean age 45 years (SD: 10); 31 female; mean age 45 years (SD: 7) Typically undertaking: 5-7 immersions/week	CWI Test 1: at start of winter swimming season): 10°C CWI Test 2: 1 month into season): 4°C CWI Test 3: 3 months into season): 4°C Mean duration of immersion 36 s. Described as a 'dip' immersion; head out	Plasma norepinephrine (nmol/l) Test 1: 4.1 (1.7) Test 2: 3.2 (1.3)*** Test 3: 3.0 (1.0) Plasma Epinephrine (nmol/l) Test 1: 0.44 (0.21) Test 2: 0.26 (0.16)** Test 3: 0.29 (0.19) Serum cortisol (nmol/l) Test 1: 573 (178) Test 2: 535 (116) Test 3: 526 (182)	Plasma norepinephrine (nmol/l) Test 1: 6.1 (2.3)*** Test 2: 6.0 *** Test 3: 5.4*** Plasma epinephrine (nmol/l) Test 1: 0.45 (0.26) Test 2: 0.31 (0.17) Test 3: 0.25 (0.17) Serum cortisol (nmol/l) Test 1: 566 (185) Test 2: 562 (151) Test 3: 562 (206)	Dip immersion therefore the duration may have been variable across participants. Levels of epinephrine showed a degree of inter subject variability. Baseline levels of norepinephrine and epinephrine decreased significantly after 1 month of winter swimming. Catecholamine response to CWI was attenuated with repeated immersions.
Dugué et al ¹⁵ (randomised controlled trial)	N=20 females Aged 20–45 years	A. Whole body cryotherapy (n=10) 2 min at –110°C (3/week for 3 months) B. CWI (n=10) 20 s in a cold pond (3/week for 3 months) Test 1: Start of study Test 2: 2 weeks Test 3: 1 month Test 4: 2 months Test 5: 3 months	TRAP (μmol/l) Test 1: A. 1108 (1027; 1190) B. 1111 (988; 1235) Test 2: A. 1113 (994; 1232) B. 1177.3 (1082; 1273) Test 3: A. 1103 (984; 1223) B. 1174 (1050; 1297) Test 4: A. 1114 (997; 1232) B. 1116 (1007; 1225) Test 5: A. 1082 (953; 1211) B. 1091 (994; 1187)	TRAP (μmol/l) Test 1: 2 min: A. +35 (15; 60); B. +6.9 (–12.5; 26.2); 35 min: A. +15 (–6; 36); B. –2.7 (–18; 13) Test 2: 2 min: A. +26 (7; 44); B. +1.8 (–18; 22) 35 min: A. –5 (–9; 9); B. –4.9 (–25; 15) Test 3: 2 min: A. +22 (8; 35); B. +29 (15; 43) 35 min: A. +10 (–11; 32); B. –4 (–31; 23) Test 4: 2 min: A. +6 (–20; 33); B. +8.9 (–15; 33) 35 min: A. –18 (–37; 1.5); B. –21 (–49; 7) Test 5: 2 min: A. +21 (–3; 44); B. +10 (2; 37) 35 min: A. +2 (–11; 15); B. –17.6 (–45; 9)	Exact temperature of water is not stated. Duration of exposure was self-timed. Values are mean and 95% CI. + Indicates an increase from pretest value. – Indicates a decrease from pretest value. Variation in baseline TRAP between participants. Regular winter swimming was not deemed to be harmful to plasma antioxidant capacity. CWI generally milder reaction than WBC.
Eglin and Tipton ¹⁶ (randomised controlled trial)	N=18 healthy participants [13 male, 5 female; Gp A: mean age 25.1 years (SD: 6); Gp B mean age 26.3 years (SD: 6.9); Gp C mean age 26 years (SD: 3.5)]	A. 10°C shower on back 3 min B. 15°C shower on back 3 min C. 10°C 30 s on back/30 s on front All groups: 6 showers over a 4-day period Test 1: at baseline Test 2: follow-up after acclimatisation. Both tests at 10°C; neck level; 3 min; swim wear	No preimmersion values provided	Inspiratory minute volume~ Test 1 0–30 s: A. 80; B. 80; C. 55 30–180 s: A. 60; B. 50 C. 45 Test 2 0–30 s: A. 78; B. 78 ; C. 55 30–180 s: A. 45; B. 35*; C. 30* Heart rate~ Test 1 0–30 s: A. 125; B. 130; C. 100 30–180 s: A. 120; B. 120 ; C. 100 Test 2 0–30 s: A. 125; B. 130 ; C. 110 30–180 s: A. 100; B. 100*; C. 95	No preimmersion values provided. Biphasic response to immersion noted: 0–30 s; 30–180 s. ~ Approximate data extracted from graph *Significant change in comparison with test 1.
Mantoni et al ²⁵ (repeated-measures design)	N=13 healthy male participants; mean age: 32 years (SD: 7); no experience of winter swimming	0°C; 30 s Slowly lowered in a sitting position; Immersion to sternal notch	Heart rate 74 (16) Minute ventilation 11 (3) P.Pco ₂ (mm/Hg) 38 (4)	Heart rate 107 (18) Minute ventilation 20 s: 66 (31) P.Pco ₂ (mm/Hg) 20 s: 26 (5) Cerebral blood-flow velocity (Middle cerebral artery) Mean decrease of 43% (8)	2/13 participants were unable to complete the test. Both these participants had the greatest reductions of cerebral blood flow, and they showed clinical signs including disorientation and clouding of consciousness.

* p<0.05, **p<0.01, ***p<0.001.

AP, alkaline phosphatase; ASAT, aspartate aminotransferase; CK, creatine kinase; CWI, cold-water immersion; γGT, γ-glutamyl transferase; HNE, 4-hydroxynonenal; P.Pco₂, partial pressure end tidal CO₂; TRAP, total radical antioxidant capacity of plasma.

Table 2 Summary of cold-water immersion protocols

Reference	Temperature (°C)	Duration (min)	Depth of immersion
Goode <i>et al</i> ¹⁷	11	Maximum 5	Neck
Malkinson <i>et al</i> ²⁴	13	4	Shoulder
Kauppinen ^{20–23}	Ice water	0.5 × 2	Neck
Tipton <i>et al</i> ²⁸	10	3	Neck
Siems <i>et al</i> ²⁶	2–3	5	Neck (swimming)
Hermanussen <i>et al</i> ¹⁸	2–6.8	Minimum 1.5	Neck
Tipton <i>et al</i> ²⁹	10	3	Neck
Siems <i>et al</i> ²⁷	1–4	Minimum 5	Neck (swimming)
Tipton <i>et al</i> ³⁰	10	3	Neck
Huttunen <i>et al</i> ¹⁹	4–10	0.5	Neck
Dugué <i>et al</i> ¹⁵	Ice water	20 s	Neck (swimming)
Eglin and Tipton ¹⁶	10	3	Neck
Mantoni <i>et al</i> ²⁵	0	0.5	Sternum

(6 × 3 min immersions in 15°C)^{29–30} or cold showering¹⁶ had an attenuation of heart-rate response during cold immersion, with some evidence of long-term habituation,³⁰ in comparison with controls.

Blood pressure

Systolic arterial pressure increased significantly ($p < 0.05$) during a 30 s immersion from 128.7 mm Hg (SD 15.4) (preimmersion) to 143.1 mm Hg (SD 17.2).²¹ This value increased further to 147.3 (14) mm Hg ($p < 0.05$) during a second immersion repeated 2 min later. In the same study, diastolic values were increased significantly ($p < 0.05$) in comparison with baseline, after the second immersion only. Of note, both systolic and diastolic arterial pressures had returned to baseline values within 30 min.

Cerebral blood flow

Mantoni *et al*²⁵ used transcranial Doppler imaging to measure the changes in blood-flow velocity in the middle cerebral artery during a 30 s immersion at 0°C. This short immersion produced a mean decrease in cerebral artery blood flow of 43% (SD 8). Of note, two out of the 13 participants were removed from the water prematurely as they showed signs of syncope characterised by drowsiness, blurred vision and a loss of responsiveness.

Respiratory response

Respiratory minute volume

Tipton *et al*²⁸ showed increases in mean minute volume from 16.4 (l/min) at baseline, to 31 and 32 l/min during the first and second minutes of CWI, respectively. Shorter immersions in lower water temperatures²⁵ resulted in larger increases, from 11 (3) l/min preimmersion, to a mean of 66 (31) l/min after 30 s of immersion. Goode *et al*¹⁷ reported values of 94.5, 71.3 and 94.6 l/min during the first three breaths taken during CWI, and Malkinson *et al*,²⁴ recorded mean values of 43.68 (SD 20.6) l/min in males and 28.74 (SD 10.35) l/min in females during a 4 min immersion. Both of these studies also noted increases in minute volume during thermo-neutral immersion at either 28°C¹⁷ or 33.18°C;²⁴ however, these changes were significantly smaller than those reported during CWI. Although the remainder of studies^{16–29–30} failed to report adequate preimmersion values, they each recorded large minute volume values, particularly during the initial stages of immersion.

End tidal carbon dioxide partial pressure (P_{CO_2})

Mantoni *et al*²⁵ found decreases from 38 (4) mm Hg preimmersion to 26 (5) mm Hg. Tipton *et al*²⁸ recorded a mean reduction of P_{CO_2} of 0.96 kPa over 3 min of CWI. Goode *et al*¹⁷ also found that alveolar P_{CO_2} dropped, from preimmersion values of 36.4 mm Hg to levels of less than 30 mm Hg, with the lowest mean value, of 23.9 mm Hg, recorded in the immediate stages of immersion.

Body temperature

The recorded rate of skin temperature reduction during immersion was between 0.301°C/s²⁸ and 0.42°C/s^{29–30} in water temperatures of 10°C, and 0.33°C/s during immersion at 15°C.²⁹ In another study,²² 30 s cold immersions reduced mean skin temperature (based on the arithmetic mean from four body sites) by 2.7°C, but had almost no effect on core temperature (−0.1°C).

Metabolism

Oxygen consumption

Mean oxygen consumption increased from 0.417 l/min preimmersion, to 0.676 l/min after 3 min of CWI.²⁸

Biochemical response

Total radical antioxidant capacity of plasma

Total radical antioxidant capacity of plasma (TRAP) values were recorded in healthy women (unaccustomed to any form of cold exposure), randomised to undertake either WBC or CWI (three times per week) over a period of 12 weeks.¹⁵ During the first 4 weeks, both groups had significant increases in TRAP values ($p < 0.05$) immediately after cold exposure, which returned to baseline levels after 35 min. Similar trends were noted in the CWI group at week 12 ($p < 0.05$), but there were no changes in participants' resting TRAP values throughout the duration of the study.

Antioxidants

There was evidence from two studies^{26–27} that uric acid levels decrease immediately after CWI. The change was significant in one of the studies²⁶ with mean values decreasing by more than half from preimmersion (310 mM (SD 126)) to 1 h postimmersion (141 mM (SD 69)), before returning to baseline within 24 h. Levels of ascorbic acid, reduced glutathione (GSH) and 4-hydroxynonenal (HNE) were also significantly decreased ($p < 0.01$)²⁷ 1 h after CWI. This corresponded with an immediate increase in the levels of oxidised glutathione (GSSH).^{26–27}

Catecholamines/cortisol

In a study by Kauppinen *et al*,²³ two short CWIs induced immediate and significant increases ($p < 0.01$) in both norepinephrine and cortisol levels. A longer 5 min immersion¹⁸ also resulted in significant increases in cortisol levels ($p < 0.05$) 30 min after immersion, which returned to resting levels after 1 h. Although Huttunen *et al*¹⁹ also reported significant increases ($p < 0.001$) in norepinephrine, they found only marginal changes in cortisol.

Biochemical status of cold acclimatised versus control

Two studies^{26–27} compared the resting levels of antioxidants in cold-acclimatised participants, to groups of controls with no experience of CWI. In both studies, the cold-acclimatised group had been undertaking regular winter swimming in temperatures of 1–4°C for between 1 and 11 years. The acclimatised participants had higher baseline levels of GSH and

total glutathione,^{26 27} catalase ($p < 0.01$)²⁷ and superoxide dismutase ($p < 0.01$).²⁷ There were no differences in glutathione peroxidase.²⁷

Response of cold-acclimatised participants to long-term CWI

Both Hermanussen *et al*¹⁸ and Huttunen *et al*¹⁹ followed groups of winter swimmers over the course of a winter swimming season lasting between 2.5¹⁸ and 3 months.¹⁹ Hermanussen *et al*¹⁸ exposed participants to at least one CWI per week for approximately 5 min, whereas Huttunen *et al*¹⁹ used short dip immersions, at more regular intervals (five to seven times per week).

Hermanussen *et al*¹⁸ found few changes to resting levels of most biochemical markers that were measured (eg, luteinising hormone; follicle-stimulating hormone; growth hormone; insulin; cortisol; thyroid stimulating hormone; prolactin and glucose); however, basal prolactin was increased, and insulin serum decreased by almost 50%. Huttunen *et al*¹⁹ revealed a trend in which resting catecholamine levels decreased significantly ($p < 0.01$) after 1 month of regular winter swimming. Moreover, while acute exposure to CWI consistently increased catecholamines levels, the magnitude of this response significantly attenuated over the 3-month study ($p < 0.01$).

DISCUSSION

This is the first systematic review to consider the biochemical and physiological effects of short durations of CWIs. Although limited to healthy participants, we have highlighted that CWI (<5 min) at temperatures lower than 15°C induce substantial alterations on numerous body systems. These include, a 'cold shock response,' extreme activation of the sympathetic nervous system, oxidative stress and a possible increase in free-radical-species formation. There is evidence that participants undertaking regular CWI have higher levels of antioxidants and that the cold shock response is attenuated with repeated immersions.

Study limitations

Before considering the wider implications of this review, it is important to consider that the vast heterogeneity of included studies and sample size limitations greatly limit the extent of our conclusions. In addition, all but one of the included studies used immersion to neck level, and potentially the results reflect the body's largest potential response to brief periods of CWI in resting participants.

Body temperature

The scientific rationale underpinning CWI for recovery after sport has yet to be determined. Clearly, CWI has a cooling effect on body tissues based on a physical heat loss to the surrounding environment. It is also associated with a peripheral vasoconstrictive response,^{31 32} reduced perfusion and a decrease in oedema formation.³³ However, inducing changes to muscle, blood flow or other deeper tissues seems to require a significant reduction to the deep tissue temperature.³⁴ None of the included studies recorded deep tissue temperatures and blood flow, and few^{22 28 29} focused on skin temperature. As expected, immediate decreases in skin temperature were reported, but this may not correlate well with deep temperature changes. Others studies have shown that despite causing rapid reductions in skin temperature, ice-pack applications (20–30 min)^{35–37} and contrast immersions^{38 39} have shown little effect on intramuscular temperature.

In certain clinical environments, the main rationale for using CWI is to reduce core temperature. In the current review, only one study²² recorded core temperature, noting decreases of just 0.1°C after 2×30 s immersions. If our aim is to reduce core body temperature, a longer CWI may be more appropriate based on this evidence. Notwithstanding this, it must be noted that this review focused specifically on resting subjects, rather than exercising or hyperthermic individuals. This may be an important area for future research in CWI. There is evidence to show that using CWIs (5–15 min durations) between running/cycling events can lower core temperature^{40–42} and muscle temperature,⁴³ and enhance subsequent performance^{40–42} in hot environmental conditions (approximately: 27–34°C air temperature; 40% relative humidity). In contrast, similar CWI protocols had a negative effect on cycling performance in non-hyperthermic conditions.^{1 4} Future studies must determine if the CWI protocol must be altered in relation to the thermal status of the athlete.

Respiratory and cardiovascular

We found evidence that CWIs were associated with a number of sudden changes to the cardiovascular and respiratory systems. This is not a new phenomenon, and collectively, these changes are often referred to as 'cold shock,' characterised by an inspiratory gasp, hyperventilation, hypnocapnia, tachycardia and hypertension.⁴⁴ This response normally reaches a peak within the first 30 s of CWI, with many individuals adapting after approximately 3 min.⁴⁵ The magnitude of cold shock is increased, during sudden or unexpected immersion,⁴⁶ or full-body immersions. One small study,²⁵ which employed a 30 s immersion at 0°C, recorded a marked reduction in cerebral blood-flow velocity, with two out of the 13 participants showing signs of imminent syncope. We also highlighted that CWIs induce an initial inspiratory gasp, large increases in minute volumes and a trend towards an increase in heart rate, blood pressure and oxygen consumption.

The effects that CWI has on the cardiovascular and respiratory systems may be short-term and clinically insignificant to the athletic population. Indeed, the levels of tachycardia seem comparable with those associated with a typical postmatch cool down. Nonetheless, it may be important to consider the mechanisms behind these changes. Some may be independent of water temperature. For example, the hydrostatic forces of water can alter the movement and distribution of body fluid increasing central blood volume, stroke volume and cardiac output.⁹ This effect may increase with immersion depth and may therefore be particularly relevant to the included studies. Two of the studies noted large increases in minute volumes, even in thermoneutral water; however, comparison with CWI showed significantly larger increases. In all cases, the respiratory and cardiac responses were immediate, perhaps suggesting that they are neurogenic in origin, relating to the extreme cutaneous stimulation associated with CWI, and a subsequent increase in sympathetic nervous activity.⁴⁵ This was perhaps also exemplified by the significant increases in norepinephrine levels observed in the current review,^{19 23} and from others using longer CWI protocols.¹⁴ Pragmatically there may be few benefits to inducing these responses immediately after exercise. Causing an increase in sympathetic activity, heart rate, blood pressure, oxygen consumption and metabolic rate does not configure with the general principles of the current sports recovery model. However, since few studies in the current review have measured longer-term outcomes, we cannot postulate if there is any associated detrimental effect.

Biochemical

To date, the biochemical effects associated with CWI in the sporting environment have been overlooked. This review shows evidence that short CWIs are associated with some degree with oxidative stress and a possible increase in free-radical-species formation. This postulation was derived predominately by assessing data showing a change in peripheral oxidative status through concentrations of systemic antioxidants^{26,27} and TRAP values.¹⁵ Free radicals are produced both as a by-product of cellular metabolism in aerobic systems and from various environmental sources. When free-radical production exceeds the capacities of protection and repair mechanisms, oxidative stress occurs, resulting in damage to macromolecules such as proteins, lipids and DNA. The relative risk of oxidative stress appears to increase during periods of elevated metabolism, including both exercise activity and cold exposure.

The mechanisms associated with free-radical production following CWI may be multifold and difficult to determine. However, there is good evidence that shivering thermoregulation is associated with free-radical production.⁴⁷ Shivering is an involuntary, repeated rhythmic muscle contraction that can cause large increases in metabolic heat production. While shivering can occur in the immediate stages of CWI,^{13,48} its onset can be delayed by approximately 40 min in acclimatised individuals such as winter swimmers.⁴⁸

Non-shivering thermogenesis can also be used for heat production and temperature regulation in cold environments. This occurs in brown adipose tissue by the uncoupling of oxidative metabolism from ATP production by uncoupling protein-1 in mitochondria, which results in the generation of heat. To sustain body temperature, an increase in metabolism is required immediately on exposure to low ambient temperatures. This elevation in metabolism may increase the production of free radicals. Therefore, it is likely that during acute cold exposure, the disparity between free-radical production and protection mechanisms should be at its greatest, which may result in a significant risk of oxidative stress.

Given that the present review focused on short CWIs, it may be more conceivable that the free-radical production and subsequent oxidative stress may have resulted from molecular oxidation. Several molecules, including catecholamines, may oxidise in the presence of oxygen to form superoxide anions. The superoxide is subsequently capable of oxidising more of the originating compound, thus setting up a complex free-radical chain reaction.⁴⁹ Given that some studies showed an increase in peripheral catecholamine concentration following cold immersion, it is possible that any increase in oxidative stress may be due to an auto-oxidative process.

Regardless of the mechanism associated with oxidative stress in the studies outlined, the overproduction of cellular and systemic free radicals can be detrimental to the proper functioning of the human body and is therefore undoubtedly an unwanted effect of ice immersion, particularly following exercise and sport. Indeed, aerobic^{50,51} and anaerobic⁵² exercise has been shown to cause an oxidative stress response on the body; therefore, inducing oxidative damage during a 'recovery' period may further exacerbate muscle stress and delay the recovery process. Moreover, should cold immersion reduce body temperature to a level that initiates non-voluntary shivering thermogenesis, the resultant increase in resting metabolism may unnecessarily cause oxidation to stored muscle glycogen and triglyceride levels, and, in doing so, further hamper the recovery process prior to exercise.

Although investigators have relied mainly on the change in systemic antioxidant status and by-products of free radical attack to lipid membranes to determine oxidative stress²⁷ (HNE), the reported data must be viewed with caution. No study used more than one marker of oxidation in order to quantify oxidative stress, and measuring two or more markers of lipid, protein or nucleic acid oxidation is recommended.⁵³ All oxidative stress-related biochemical indices utilised are considered indirect markers or 'foot prints' of oxidative stress. It is recommended that future studies in this area include a more direct measurement of free-radical production, such as electron paramagnetic resonance spectroscopy, in order to quantify more accurately the level of potential oxidative damage caused by cold immersion.

Sporadic or unaccustomed levels of exercise are associated with high levels of free-radical damage.⁵⁴ Similarly, we must ascertain if athletes undertaking sporadic CWIs are at similar risk. The long-term effect of CWI in the athletic population must also be fully elucidated. Our current evidence base is limited to profiling the status of winter swimmers, individuals who regularly undertake short CWIs or swimming, in cold temperatures.⁵⁵ This practice has been associated with a 'hardening' response to oxidative stress, and a resultant protective effect against free-radical-induced tissue damage.²⁷ This may explain some of the perceived benefits reported by athletes during recovery from exercise, particularly those at an elite level undertaking regular CWI.

Precautions

Few practitioners and athletes may be aware of the associated systemic effects we have highlighted in this review. A number of these effects may require further study; however, until this time, we would recommend a judicious approach in certain clinical situations. The changes to heart rate and blood pressure may be clinically unremarkable, but it must be noted that others have reported cardiac arrhythmias,^{56,57} and increased QT dispersion⁵⁸ with cold-water submersions and extreme full body immersions. Equally, while the risk of hyperventilation and loss of consciousness are restricted to full body immersion in temperatures of 0°C, it may still be judicious for athletes, particularly those who are unaccustomed to CWI, to have adequate supervision and potentially undertake a period of acclimatisation.

CONCLUSION

Based on the current review, the physiological and biochemical rationale for using CWI in sports recovery still remains largely unclear. Many more basic sciences and clinical studies are merited in order to develop evidence-based guidelines on the safe and effective practice of CWI for the entire athletic population.

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Detail has been removed from this case description/these case descriptions to ensure anonymity. The editors and reviewers have seen the detailed information available and are satisfied that the information backs up the case the authors are making.

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